

Cumulants

Definition 1.1. The **moment generating function** of a random variable X is:

$$M_X(s) = \mathbb{E}(e^{Xs}) = \int e^{xs} f(x) dx.$$

The *moments* of X are the coefficients m_j in the expansion:

$$M_X(s) = \sum_{j=0}^{\infty} m_j \frac{s^j}{j!} = 1 + m_1 s + m_2 \frac{s^2}{2!} + m_3 \frac{s^3}{3!} + \dots$$

The **cumulant generating function** is the logarithm of the MGF:

$$K_X(s) = \ln M_X(s).$$

The *cumulants* of X are defined by the coefficients κ_j in the expansion:

$$K_X(s) = \sum_{j=1}^{\infty} \kappa_j \frac{s^j}{j!} = \kappa_1 s + \kappa_2 \frac{s^2}{2!} + \kappa_3 \frac{s^3}{3!} + \dots$$

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Example 1.1. Cumulants are related to moments via the identity:

$$\begin{aligned} \sum_{j=1}^{\infty} \kappa_j \frac{s^j}{j!} &= K_X(s) = \ln M_X(s) = \ln \left(\sum_{j=0}^{\infty} m_j \frac{s^j}{j!} \right) = \ln \left(1 + \sum_{j=1}^{\infty} m_j \frac{s^j}{j!} \right) \\ &= \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} \left(\sum_{j=1}^{\infty} m_j \frac{s^j}{j!} \right)^k. \end{aligned}$$

Expanding the sums gives:

$$\begin{aligned} \kappa_1 s + \kappa_2 \frac{s^2}{2!} + \kappa_3 \frac{s^3}{3!} + \dots &= + \frac{1}{1} \left(m_1 s + m_2 \frac{s^2}{2!} + m_3 \frac{s^3}{3!} + m_4 \frac{s^4}{4!} + \dots \right) \\ &\quad - \frac{1}{2} \left(m_1 s + m_2 \frac{s^2}{2!} + m_3 \frac{s^3}{3!} + \dots \right)^2 \\ &\quad + \frac{1}{3} \left(m_1 s + m_2 \frac{s^2}{2!} + \dots \right)^3 - \frac{1}{4} (m_1 s + \dots)^4 + \dots \end{aligned}$$

Equating coefficients gives:

$$\begin{aligned} \kappa_1 &= m_1, \\ \kappa_2 &= m_2 - m_1^2, \\ \kappa_3 &= m_3 - 3m_1 m_2 + 2m_1^3, \\ \kappa_4 &= m_4 - 4m_1 m_3 - 3m_2^2 + 12m_1^2 m_2 - 6m_1^4. \end{aligned}$$

Alternatively, the moments are related to the cumulants via the identity:

$$\begin{aligned} \sum_{j=0}^{\infty} m_j \frac{s^j}{j!} &= M_X(s) = \exp \{K_X(s)\} = \exp \left\{ \sum_{j=1}^{\infty} \kappa_j \frac{s^j}{j!} \right\} = \prod_{j=1}^{\infty} \exp \left\{ \kappa_j \frac{s^j}{j!} \right\} \\ &= \prod_{j=1}^{\infty} \left\{ \sum_{l=0}^{\infty} \frac{1}{l!} \left(\kappa_j \frac{s^j}{j!} \right)^l \right\} = \prod_{j=1}^{\infty} \left\{ 1 + \kappa_j \frac{s^j}{j!} + \kappa_j^2 \frac{s^{2j}}{2!(j!)^2} + \cdots \right\}. \end{aligned}$$

Expanding the sum and product gives:

$$\begin{aligned} 1 + m_1 s + m_2 \frac{s^2}{2!} + m_3 \frac{s^3}{3!} + m_4 \frac{s^4}{4!} + \cdots &= \left\{ 1 + \kappa_1 s + \kappa_1^2 \frac{s^2}{2!} + \kappa_1^3 \frac{s^3}{3!} + \kappa_1^4 \frac{s^4}{4!} + \cdots \right\} \\ &\times \left\{ 1 + \kappa_2 \frac{s^2}{2!} + \kappa_2^2 \frac{s^4}{2!(2!)^2} + \cdots \right\} \\ &\times \left\{ 1 + \kappa_3 \frac{s^3}{3!} + \cdots \right\} \times \left\{ 1 + \kappa_4 \frac{s^4}{4!} + \cdots \right\} \times \cdots \end{aligned}$$

Equating coefficients gives:

$$\begin{aligned} m_1 &= \kappa_1, \\ m_2 &= \kappa_1^2 + \kappa_2, \\ m_3 &= \kappa_1^3 + 3\kappa_1\kappa_2 + \kappa_3, \\ m_4 &= \kappa_1^4 + 6\kappa_1^2\kappa_2 + 3\kappa_2^2 + 4\kappa_1\kappa_3 + \kappa_4. \end{aligned}$$



Hermite Polynomials

2.1 Basic Properties

Definition 2.1. The **Hermite polynomials** $H_k(x)$ are defined by the series expansion of the generating function $G_H(x, t)$:

$$G_H(x, t) = e^{xt-t^2/2} = \sum_{k=0}^{\infty} H_k(x) \frac{t^k}{k!}. \quad (2.1)$$

The first several Hermite polynomials are:

$$\begin{aligned} H_1(z) &= z, \\ H_2(z) &= z^2 - 1, \\ H_3(z) &= z^3 - 3z, \\ H_4(z) &= z^4 - 6z^2 + 3. \end{aligned}$$

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Proposition 2.1. The Hermite polynomials satisfy the recursion:

$$\dot{H}_k(x) = kH_{k-1}(x) \text{ for } k \geq 1.$$

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Proof. Differentiating the exponential representation of (2.1) w.r.t. x :

$$\begin{aligned} \frac{\partial}{\partial x} G_H(x, t) &= \frac{\partial}{\partial x} e^{xt-t^2/2} = t e^{xt-t^2/2} = t G_H(x, t) \\ &= \sum_{k=0}^{\infty} H_k(x) \frac{t^{k+1}}{k!} = \sum_{k=0}^{\infty} (k+1) H_k(x) \frac{t^{k+1}}{(k+1)!} = \sum_{k=1}^{\infty} k H_{k-1}(x) \frac{t^k}{k!}. \end{aligned}$$

Differentiating the series representation of (2.1) w.r.t. x :

$$\frac{\partial}{\partial x} G_H(x, t) = \frac{\partial}{\partial x} \sum_{k=0}^{\infty} H_k(x) \frac{t^k}{k!} = \sum_{k=0}^{\infty} \dot{H}_k(x) \frac{t^k}{k!} = \dot{H}_0(x) + \sum_{k=1}^{\infty} \dot{H}_k(x) \frac{t^k}{k!}$$

Equating powers of t gives:

$$\dot{H}_k(x) = kH_{k-1}(x) \text{ for } k \geq 1.$$

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Proposition 2.2. The Hermite polynomials satisfy the recursion:

$$H_{k+1}(x) = xH_k(x) - kH_{k-1}(x) \text{ for } k \geq 1.$$

◆

Proof. Differentiating the exponential representation of (2.1) w.r.t. t :

$$\begin{aligned}\frac{\partial}{\partial t}G_H(x, t) &= \frac{\partial}{\partial t}e^{xt-t^2/2} = (x-t)e^{xt-t^2/2} = (x-t)G_H(x, t) \\ &= \sum_{k=0}^{\infty} xH_k(x)\frac{t^k}{k!} - \sum_{k=0}^{\infty} (k+1)H_k(x)\frac{t^{k+1}}{(k+1)!} \\ &= \sum_{k=0}^{\infty} xH_k(x)\frac{t^k}{k!} - \sum_{k=0}^{\infty} kH_{k-1}(x)\frac{t^k}{k!}.\end{aligned}$$

Differentiating the series representation of (2.1) w.r.t. t :

$$\frac{\partial}{\partial t}G_H(x, t) = \frac{\partial}{\partial t} \sum_{k=0}^{\infty} H_k(x)\frac{t^k}{k!} = \sum_{k=1}^{\infty} H_k(x)\frac{t^{k-1}}{(k-1)!} = \sum_{k=0}^{\infty} H_{k+1}(x)\frac{t^k}{k!}$$

Equating powers of t gives:

$$H_{k+1}(x) = xH_k(x) - kH_{k-1}(x).$$

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Result 2.1 (Rodrigues' Formula). The Hermite polynomials satisfy:

$$(-1)^k \frac{d^k}{dx^k} \phi(x) = H_k(x)\phi(x), \quad (2.2)$$

where $\phi(x)$ is the standard normal density.

♣

Proof. The Hermite generating function (2.1) is expressible as:

$$G_H(x, t) = e^{xt-t^2/2} = e^{x^2/2-(t-x)^2/2} = \sum_{k=0}^{\infty} H_k(x)\frac{t^k}{k!}.$$

The k th Hermite polynomial is given by:

$$\left\{ \frac{\partial^k}{\partial t^k} e^{xt-t^2/2} \right\}_{t=0} = \left\{ \frac{\partial^k}{\partial t^k} \sum_{l=0}^{\infty} H_l(x)\frac{t^l}{l!} \right\}_{t=0} = \left\{ \sum_{l=k}^{\infty} H_l(x)\frac{t^{l-k}}{(l-k)!} \right\}_{t=0} = H_k(x).$$

Replacing the kernel of the generating function:

$$H_k(x) = \left\{ \frac{\partial^k}{\partial t^k} e^{xt-t^2/2} \right\}_{t=0} = \left\{ \frac{\partial^k}{\partial t^k} e^{x^2/2-(t-x)^2/2} \right\}_{t=0} = e^{x^2/2} \left\{ \frac{\partial^k}{\partial t^k} e^{-(t-x)^2/2} \right\}_{t=0}.$$

Observe that:

$$\frac{\partial}{\partial t} e^{-(t-x)^2/2} = -(t-x)e^{-(t-x)^2/2} = -\frac{\partial}{\partial x} e^{-(t-x)^2/2}.$$

Replacing the derivatives in t with derivatives in x :

$$\begin{aligned} H_k(x) &= e^{x^2/2} \left\{ \frac{\partial^k}{\partial t^k} e^{-(t-x)^2/2} \right\}_{t=0} \\ &= e^{x^2/2} \left\{ (-1)^k \frac{\partial^k}{\partial x^k} e^{-(t-x)^2/2} \right\}_{t=0} = e^{x^2/2} (-1)^k \frac{d^k}{dx^k} e^{-x^2/2}. \end{aligned}$$

Multiplying through by $\phi(x) = (2\pi)^{-1/2} \exp(-x^2/2)$ gives:

$$\phi(x) H_k(x) = (-1)^k \frac{d^k}{dx^k} \phi(x).$$

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Result 2.2 (Orthogonality). The Hermite polynomials form an orthogonal system to the ϕ -weighted inner product:

$$\langle H_m, H_n \rangle_\phi = \int_{-\infty}^{\infty} H_m(x) H_n(x) \phi(x) dx = \begin{cases} n!, & n = m, \\ 0, & n \neq m. \end{cases}$$

♣

Proof. Multiply two instances of the Hermite generating function (2.1):

$$G_H(x, s) G_H(x, t) = e^{xs-s^2/2} e^{xt-t^2/2} = \sum_{m=0}^{\infty} \sum_{n=0}^{\infty} H_m(x) H_n(x) \frac{s^m t^n}{m! n!}.$$

Multiplying the series representation by $e^{-x^2/2}$ and integrating gives:

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{s^m t^n}{m! n!} \int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2/2} dx.$$

Multiplying the exponential representation by $e^{-x^2/2}$ and integrating gives:

$$\int_{-\infty}^{\infty} e^{xs-s^2/2+xt-t^2/2-x^2/2} dx = e^{st} \int_{-\infty}^{\infty} e^{-(x-s-t)^2/2} dx = e^{st} \sqrt{2\pi} = \sqrt{2\pi} \sum_{k=0}^{\infty} \frac{(st)^k}{k!}$$

Since the two representations are equivalent:

$$\sum_{m=0}^{\infty} \sum_{n=0}^{\infty} \frac{s^m t^n}{m! n!} \int_{-\infty}^{\infty} H_m(x) H_n(x) e^{-x^2/2} dx = \sqrt{2\pi} \sum_{k=0}^{\infty} \frac{(st)^k}{k!}$$

Equating coefficients gives the result. ■

Fourier Transform

Definition 3.1. Define the **Fourier transform** as:

$$\mathcal{F}\{g(x)\} = \int_{-\infty}^{\infty} g(x)e^{-i\omega x} dx,$$

and the **inverse Fourier transform** as:

$$\mathcal{F}^{-1}\{G(\omega)\} = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega)e^{i\omega x} d\omega.$$

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Result 3.1. $\phi(t)$ is an **eigenfunction** of the Fourier transform and its inverse:

$$\begin{aligned}\mathcal{F}\{\phi(x)\} &= e^{-\omega^2/2} = \sqrt{2\pi}\phi(\omega), \\ \mathcal{F}^{-1}\{\sqrt{2\pi}\phi(\omega)\} &= \mathcal{F}^{-1}\{e^{-\omega^2/2}\} = \phi(x).\end{aligned}$$

♣

Proof. The Fourier transform of $\phi(x)$ is:

$$\begin{aligned}\mathcal{F}\{\phi(x)\} &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(x+i\omega)^2/2 - \omega^2/2} dx \\ &= \frac{1}{\sqrt{2\pi}} e^{-\omega^2/2} \int_{-\infty}^{\infty} e^{-u^2/2} du = e^{-\omega^2/2}.\end{aligned}$$

The inverse Fourier transform of $e^{-\omega^2/2}$ is:

$$\begin{aligned}\mathcal{F}^{-1}(e^{-\omega^2/2}) &= \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\omega^2/2} e^{i\omega x} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-(\omega-ix)^2/2 - x^2/2} d\omega \\ &= \frac{1}{2\pi} e^{-x^2/2} \int_{-\infty}^{\infty} e^{-u^2/2} du = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.\end{aligned}$$

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Corollary 3.1. The Fourier transform of $x^k\phi(x)$ is:

$$\mathcal{F}\{x^k\phi(x)\} = i^k(-1)^k H_k(\omega) e^{-\omega^2/2}.$$

The inverse Fourier transform of $\omega^k e^{-\omega^2/2}$ is:

$$\mathcal{F}^{-1}\{\omega^k e^{-\omega^2/2}\} = i^k H_k(x) \phi(x).$$

♣

Proof. Observe that:

$$\frac{\partial}{\partial \omega} e^{-i\omega x} = (-i)x e^{-i\omega x}, \quad x e^{-i\omega x} = \frac{1}{(-i)} \frac{\partial}{\partial \omega} e^{-i\omega x}, \quad x^k e^{-i\omega x} = \frac{1}{(-i)^k} \frac{\partial^k}{\partial \omega^k} e^{-i\omega x}.$$

The Fourier transform of $x^k \phi(x)$ is:

$$\begin{aligned} \mathcal{F}\{x^k \phi(x)\} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} x^k e^{-i\omega x} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/2} \frac{1}{(-i)^k} \frac{\partial^k}{\partial \omega^k} e^{-i\omega x} dx \\ &= i^k \frac{d^k}{d\omega^k} \mathcal{F}\{\phi(x)\} = i^k \frac{d^k}{d\omega^k} e^{-\omega^2/2} = i^k (-1)^k H_k(\omega) e^{-\omega^2/2}. \end{aligned}$$

Similarly:

$$\begin{aligned} \mathcal{F}^{-1}\{\omega^k e^{-\omega^2/2}\} &= \frac{1}{2\pi} \int_{-\infty}^{\infty} \omega^k e^{-\omega^2/2} e^{i\omega x} d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{1}{i^k} \frac{\partial^k}{\partial x^k} e^{i\omega x} dx \\ &= (-i)^k \frac{d^k}{dx^k} \mathcal{F}^{-1}\{e^{-\omega^2/2}\} = (-i)^k \frac{d^k}{dx^k} \phi(x) = i^k H_k(x) \phi(x). \end{aligned}$$

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3.1 Fourier-Hermite Expansion

Theorem 3.1. The Hermite polynomials form a *complete*, orthogonal basis for the Hilbert space $\mathcal{H}$ of square-integrable functions w.r.t. to the ϕ -weighted inner product:

$$\langle f, g \rangle_{\phi} = \int_{-\infty}^{\infty} f(x) g(x) \phi(x) dx.$$

□

Corollary 3.2. Any $h \in \mathcal{H}$ admits a **Fourier-Hermite expansion**:

$$h(x) = \sum_{k=0}^{\infty} \alpha_k H_k(x).$$

To determine the coefficients (α_k) , take the inner product of h wrt the m th Hermite polynomial H_m :

$$\begin{aligned} \langle h, H_m \rangle_{\phi} &= \sum_{k=0}^{\infty} \alpha_k \langle H_k, H_m \rangle_{\phi} = \alpha_m m!, \\ \alpha_m &= \frac{1}{m!} \langle h, H_m \rangle_{\phi} = \frac{1}{m!} \int_{-\infty}^{\infty} h(x) H_m(x) \phi(x) dx. \end{aligned}$$

Alternatively, consider the Hilbert space $\mathcal{L}^2$ of square integrable functions with the inner product:

$$\langle f, g \rangle = \int_{-\infty}^{\infty} f(x) g(x) dx.$$

The **Gram-Charlier expansion** with the normal kernel takes the form:

$$h(x) = \sum_{k=0}^{\infty} \alpha_k H_k(x) \phi(x). \quad (3.1)$$

Suppose $X \sim f_X$ with $f_X \in \mathcal{L}^2$. The coefficients of the *Gram-Charlier expansion* are:

$$\begin{aligned} \langle f, H_m \rangle &= \sum_{k=0}^{\infty} \alpha_k \langle H_k \phi, H_m \rangle = \alpha_m m!, \\ \alpha_m &= \frac{1}{m!} \langle h, H_m \rangle = \frac{1}{m!} \int_{-\infty}^{\infty} f(x) H_m(x) dx = \frac{1}{m!} \mathbb{E}_X \{H_m(X)\}. \end{aligned}$$



Edgeworth Expansion

Definition 4.1. The **characteristic function** of a random variable X is the scaled inverse Fourier transform of the density f_X :

$$\psi_X(\omega) = \int_{-\infty}^{\infty} f_X(x) e^{i\omega x} dx.$$

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Theorem 4.1 (Characteristic Function Inversion). The density f_X is recovered from the characteristic function $\psi(\omega)$ via:

$$f_X(x) = \frac{1}{2\pi} \int \psi_X(\omega) e^{-i\omega x} d\omega. \quad (4.1)$$

□

Proposition 4.1. Suppose $X_i \stackrel{\text{iid}}{\sim} F_X$. Define the standardized observation Z_i and the standardized sum S :

$$Z_i = \frac{(X_i - \mu)}{\sigma}, \quad S = \sqrt{n} \frac{(\bar{X} - \mu)}{\sigma}.$$

The CGFs of S and Z are related via:

$$K_S(t) = nK_Z\left(\frac{t}{\sqrt{n}}\right).$$

◆

Proof. The MGFs are related by:

$$\begin{aligned} M_S(t) &= \mathbb{E}\{e^{tS}\} = \mathbb{E}\left[\exp\left\{t \cdot \frac{\sqrt{n}}{\sigma}(\bar{X} - \mu)\right\}\right] = \mathbb{E}\left[\exp\left\{t \cdot \frac{\sqrt{n}}{\sigma} \left(\frac{1}{n} \sum_{i=1}^n X_i - \mu\right)\right\}\right] \\ &= \mathbb{E}\left[\exp\left\{t \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{(X_i - \mu)}{\sigma}\right\}\right] = \mathbb{E}\left[\exp\left\{t \cdot \frac{1}{\sqrt{n}} \sum_{i=1}^n Z_i\right\}\right] \\ &= \mathbb{E}\left[\prod_{i=1}^n e^{\frac{t}{\sqrt{n}} Z_i}\right] = \prod_{i=1}^n \mathbb{E}\left\{e^{\frac{t}{\sqrt{n}} Z_i}\right\} = \left\{M_Z\left(\frac{t}{\sqrt{n}}\right)\right\}^n. \end{aligned}$$

Taking the logarithm gives the result. ■

Proposition 4.2. A formal expansion for the CDF of S is given by:

$$F_S(s) = \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2 - i\omega t} \left\{ 1 + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\sum_{j=3}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2-1}} \right)^k \right\} d\omega dt, \quad (4.2)$$

where κ_j is the j th cumulant of the standardized observation Z_i . ◆

Proof. From the inversion identity in (4.1):

$$F_S(s) = \int_{-\infty}^s f_S(t) dt = \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} \psi_S(\omega) e^{-i\omega t} d\omega dt,$$

where $\psi_S(\omega) = M_S(i\omega)$ is the characteristic function of S .

Expressing $\psi_S(\omega)$ in terms of the CGF for Z :

$$\begin{aligned} F_S(s) &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} \left\{ M_Z \left(\frac{i\omega}{\sqrt{n}} \right) \right\}^n e^{-i\omega t} d\omega \\ &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} \exp \left\{ n K_Z \left(\frac{i\omega}{\sqrt{n}} \right) - i\omega t \right\} d\omega dt. \end{aligned}$$

Expanding the CGF as a power series:

$$F_S(s) = \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} \exp \left\{ n \sum_{j=1}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2}} - i\omega t \right\} d\omega dt.$$

Since Z is standardized $\kappa_1 = 0$ and $\kappa_2 = 1$:

$$\begin{aligned} F_S(s) &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} \exp \left\{ n \left(\frac{(i\omega)^2}{2!n} + \sum_{j=3}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2}} \right) - i\omega t \right\} d\omega dt \\ &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} \exp \left\{ -\frac{\omega^2}{2} - i\omega t + \sum_{j=3}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2-1}} \right\} d\omega dt \\ &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2 - i\omega t} \exp \left\{ \sum_{j=3}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2-1}} \right\} d\omega dt. \end{aligned}$$

Taylor expanding the exponential term that depends on n gives:

$$F_S(s) = \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2 - i\omega t} \left\{ 1 + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\sum_{j=3}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2-1}} \right)^k \right\} d\omega dt.$$

■

Corollary 4.1. A formal expansion for the density of S is given by:

$$f_S(s) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-\omega^2/2 - i\omega s} \left\{ 1 + \sum_{k=1}^{\infty} \frac{1}{k!} \left(\sum_{j=3}^{\infty} \kappa_j \frac{(i\omega)^j}{j! n^{j/2-1}} \right)^k \right\} d\omega. \quad (4.3)$$

♣

Remark 4.1. The **Edgeworth expansion** for the standardized sum S is obtained by evaluating the integrand of (4.2) term-wise. ♦

Proposition 4.3. The $\mathcal{O}(n^{-3/2})$ Edgeworth expansion for $F(s)$ is:

$$F_S(s) = \Phi(s) - \phi(s) \left\{ \frac{\kappa_3}{6n^{1/2}} H_2(s) + \frac{\kappa_4}{24n} H_3(s) + \frac{\kappa_3^2}{72n} H_5(s) \right\} + \mathcal{O}(n^{-3/2}).$$

◆

Proof. The series from (4.2) is expressible as:

$$\left\{ 1 + \left(\kappa_3 \frac{(i\omega)^3}{3!n^{1/2}} + \kappa_4 \frac{(i\omega)^4}{4!n} + \cdots \right) + \frac{1}{2} \left(\kappa_3 \frac{(i\omega)^3}{3!n^{1/2}} + \cdots \right)^2 + \cdots \right\}.$$

The $\mathcal{O}(1)$ term of the expansion is:

$$I_0 = \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2 - i\omega t} d\omega dt.$$

By completing the square:

$$e^{-\omega^2/2 - i\omega t} = e^{-t^2/2 - (\omega + it)^2/2}. \quad (4.4)$$

Evaluating the $\mathcal{O}(1)$ term:

$$\begin{aligned} I_0 &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-t^2/2 - (\omega + it)^2/2} d\omega dt \\ &= \frac{1}{2\pi} \int_{-\infty}^s e^{-t^2/2} \int_{-\infty}^{\infty} e^{-(\omega + it)^2/2} d\omega dt = \int_{-\infty}^s \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt = \Phi(s). \end{aligned}$$

The $\mathcal{O}(n^{-1/2})$ term is:

$$\frac{\kappa_3}{3!n^{1/2}} I_1 \equiv \frac{\kappa_3}{3!n^{1/2} \cdot 2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2} (i\omega)^3 e^{-i\omega t} d\omega dt.$$

Observe that:

$$(i\omega) e^{-i\omega t} = (-1) \frac{\partial}{\partial t} e^{-i\omega t}, \quad (i\omega)^k e^{-i\omega t} = (-1)^k \frac{\partial^k}{\partial t^k} e^{-i\omega t}.$$

The integral in the $\mathcal{O}(n^{-1/2})$ term reduces to:

$$\begin{aligned} I_1 &= \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2} (-1)^3 \frac{\partial^3}{\partial t^3} e^{-i\omega t} d\omega dt = \frac{1}{2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} (-1)^3 \frac{\partial^3}{\partial t^3} e^{-\omega^2/2 - i\omega t} d\omega dt \\ &= \frac{1}{2\pi} \int_{-\infty}^s (-1)^3 \frac{d^3}{dt^3} \int_{-\infty}^{\infty} e^{-\omega^2/2 - i\omega t} d\omega dt = \frac{1}{2\pi} \int_{-\infty}^s (-1)^3 \frac{d^3}{dt^3} \int_{-\infty}^{\infty} e^{-t^2/2 - (\omega + it)^2/2} d\omega dt \\ &= \frac{1}{2\pi} \int_{-\infty}^s (-1)^3 \frac{d^3}{dt^3} e^{-t^2/2} \int_{-\infty}^{\infty} e^{-(\omega + it)^2/2} d\omega dt = \int_{-\infty}^s (-1)^3 \frac{d^3}{dt^3} \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt \\ &= \int_{-\infty}^s (-1)^3 \frac{d^3}{dt^3} \phi(t) dt = (-1)^3 \int_{-\infty}^s \frac{d^3}{dt^3} \phi(t) dt = (-1)^3 \frac{d^2}{ds^2} \phi(s). \end{aligned}$$

Applying Rodrigues' formula (2.1):

$$I_1 = (-1)(-1)^2 \frac{d^2}{ds^2} \phi(s) = (-1)H_2(s)\phi(s).$$

Combining the $\mathcal{O}(1)$ and $\mathcal{O}(n^{-1/2})$ terms of the Edgeworth expansion:

$$F_S(s) = \Phi(s) - \frac{\kappa_3}{6n^{1/2}} H_2(s)\phi(s) + \mathcal{O}(n^{-1}). \quad (4.5)$$

Obtaining the $\mathcal{O}(n^{-1})$ term of the Edgeworth series requires consideration of the $k = 2$ term from (4.2). The contribution of the $k = 1$ order term from (4.2) is:

$$\frac{\kappa_4}{4!n \cdot 2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2} (i\omega)^4 e^{-i\omega t} d\omega dt = -\frac{\kappa_4}{24n} H_3(s)\phi(s),$$

where evaluation of the integral follows the same procedure as for the $\mathcal{O}(n^{-1/2})$ term. The contribution of the $k = 2$ term from (4.2) is:

$$\frac{\kappa_3^2}{2!(3!n^{1/2})^2 \cdot 2\pi} \int_{-\infty}^s \int_{-\infty}^{\infty} e^{-\omega^2/2} (i\omega)^6 e^{-i\omega t} d\omega dt = -\frac{\kappa_3^2}{72n} H_5(s)\phi(s).$$

Combining the $\mathcal{O}(n^{-1})$ terms with (4.5) gives the result. ■

Corollary 4.2. An analogous derivation applied to (4.3) gives:

$$f_S(s) = \phi(s) \left\{ 1 + \frac{\kappa_3}{6n^{1/2}} H_3(s) + \frac{\kappa_4}{24n} H_4(s) + \frac{\kappa_3^2}{72n} H_6(s) \right\} + \mathcal{O}(n^{-3/2}).$$

♣